

ANALYTIC ASPECTS OF QUADRATIC FORMS (CTNT 2026)

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1. INTRODUCTION

A *quadratic form* is a polynomial that is homogeneous of degree 2, such as

$$x^2 + y^2, \quad 5x^2 - 14xy + 10y^2, \quad \frac{7}{3}xy + \frac{15}{8}xz - \frac{12}{5}yz, \quad x_1^2 + x_2^2 + \cdots + x_k^2.$$

The first two examples above depend on two variables, so they are called binary. The third example depends on three variables and is called ternary. A quadratic form in k variables, called a k -dimensional quadratic form, will be written as $Q(x_1, \dots, x_k)$ or $Q(\mathbf{x})$ when $\mathbf{x}$ is understood to mean $(x_1, \dots, x_k)$. We will focus on quadratic forms with coefficients in $\mathbf{Z}$, which are called *integral* quadratic forms. Similarly, *rational* quadratic forms have coefficients in $\mathbf{Q}$.

A lot of classical number theory is concerned with the behavior of integral quadratic forms. Here are three such results:

- describe primes of the form $x^2 + y^2$ where $x, y \in \mathbf{Z}$ (Fermat's two-square theorem),
- describe the integral solutions of $x^2 - dy^2 = 1$ for $d \in \mathbf{Z}$ that is not a perfect square (Pell's equation),
- show every nonnegative integer is $a^2 + b^2 + c^2 + d^2$ for some $a, b, c, d \in \mathbf{Z}$ (Lagrange's four-square theorem).¹

The simplest k -dimensional quadratic form is $x_1^2 + x_2^2 + \cdots + x_k^2$, which is the dot product $\mathbf{x} \cdot \mathbf{x}$. When a k -dimensional quadratic form has coefficients in a field *not* of characteristic 2, we can write it in terms of the dot product and a $k \times k$ symmetric matrix, as follows.

Theorem 1.1. *Over a field of characteristic not equal to 2, a k -dimensional quadratic form $Q(\mathbf{x})$ is $\mathbf{x} \cdot A\mathbf{x}$ for a unique $k \times k$ symmetric matrix A .*

Proof. We have

$$\begin{aligned} Q(x_1, \dots, x_k) &= \sum_{i=1}^k a_i x_i^2 + \sum_{1 \leq i < j \leq k} a_{ij} x_i x_j \\ &= \sum_{i=1}^k a_i x_i^2 + \sum_{1 \leq i < j \leq k} \left(\frac{a_{ij}}{2} x_i x_j + \frac{a_{ij}}{2} x_j x_i \right) \\ (1.1) \quad &= \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix} \cdot \begin{pmatrix} a_1 & a_{12}/2 & \cdots & a_{1k}/2 \\ a_{12}/2 & a_2 & \cdots & a_{2k}/2 \\ \vdots & \vdots & \ddots & \vdots \\ a_{1k}/2 & a_{2k}/2 & \cdots & a_k \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{pmatrix}, \end{aligned}$$

¹Section 6 of <https://kconrad.math.uconn.edu/blurbs/ugradnumthy/descent.pdf> has a proof of the four-square theorem by descent.

which is $\mathbf{x} \cdot A\mathbf{x}$ where A is the symmetric matrix in (1.1): the i th diagonal entry of A is the coefficient in Q of x_i^2 and the (i, j) -entry of A is half the coefficient in Q of $x_i x_j$.

Why is there just one $k \times k$ symmetric matrix A such that $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ for all $\mathbf{x}$? Suppose $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ and $Q(\mathbf{x}) = \mathbf{x} \cdot A'\mathbf{x}$ for all $\mathbf{x}$, where A and A' are both $k \times k$ symmetric matrices, so $\mathbf{x} \cdot A\mathbf{x} = \mathbf{x} \cdot A'\mathbf{x}$ for all $\mathbf{x}$. We want to show $A = A'$ and will use the trick of replacing $\mathbf{x}$ with $\mathbf{x} + \mathbf{y}$: $(\mathbf{x} + \mathbf{y}) \cdot A(\mathbf{x} + \mathbf{y}) = (\mathbf{x} + \mathbf{y}) \cdot A'(\mathbf{x} + \mathbf{y})$ for all $\mathbf{x}$ and $\mathbf{y}$, and expanding both sides gives us

$$\mathbf{x} \cdot A\mathbf{x} + \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} + \mathbf{y} \cdot A\mathbf{y} = \mathbf{x} \cdot A'\mathbf{x} + \mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x} + \mathbf{y} \cdot A'\mathbf{y}.$$

Canceling the equal first terms and the equal last terms on both sides,

$$(1.2) \quad \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} = \mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x}$$

A *fundamental* property links the dot product and matrix transpose: for $k \times k$ matrices M , $M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M^\top \mathbf{y}$. Thus when M is symmetric, so $M^\top = M$, we get $M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M\mathbf{y}$: a symmetric matrix can be moved to the other side of a dot product. Apply this to (1.2): since $A^\top = A$,

$$\mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} = \mathbf{x} \cdot A\mathbf{y} + A\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot A\mathbf{y} + \mathbf{x} \cdot A\mathbf{y} = 2\mathbf{x} \cdot A\mathbf{y}.$$

Similarly, $\mathbf{x} \cdot A'\mathbf{y} + \mathbf{y} \cdot A'\mathbf{x} = 2\mathbf{x} \cdot A'\mathbf{y}$, so $2\mathbf{x} \cdot A\mathbf{y} = 2\mathbf{x} \cdot A'\mathbf{y}$ and thus $\mathbf{x} \cdot A\mathbf{y} = \mathbf{x} \cdot A'\mathbf{y}$. This equation holds for all $\mathbf{x}$ and $\mathbf{y}$, so letting $\mathbf{x}$ and $\mathbf{y}$ run through the standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_k$, we get $\mathbf{e}_i \cdot A\mathbf{e}_j = \mathbf{e}_i \cdot A'\mathbf{e}_j$. That says A and A' have the same (i, j) -component for all i and j , so $A = A'$. $\square$

Example 1.2. Taking $k = 2$, $3x^2 - xy + 5y^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} 3 & -1/2 \\ -1/2 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

Example 1.3. More generally, $ax^2 + bxy + cy^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$.

Example 1.4. In the ternary case,

$$xy + xz + yz = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \cdot \begin{pmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

When we write a quadratic form $Q(\mathbf{x})$ as $\mathbf{x} \cdot A\mathbf{x}$ with A being a symmetric matrix, set $B(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot A\mathbf{y}$. This is a bilinear form: $B(\mathbf{x}, \mathbf{y})$ is linear in $\mathbf{x}$ when $\mathbf{y}$ is fixed and it is linear in $\mathbf{y}$ when $\mathbf{x}$ is fixed.² It is also symmetric in $\mathbf{x}$ and $\mathbf{y}$ since $\mathbf{x} \cdot A\mathbf{y} = A^\top \mathbf{x} \cdot \mathbf{y} = A\mathbf{x} \cdot \mathbf{y} = \mathbf{y} \cdot A\mathbf{x} = B(\mathbf{y}, \mathbf{x})$.

Example 1.5. If $Q(\mathbf{x}) = x_1^2 + x_2^2 + \dots + x_k^2 = \mathbf{x} \cdot \mathbf{x}$, then $B(\mathbf{x}, \mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$.

Example 1.6. If $Q(x, x') = 5x^2 - 14xx' + 10x'^2 = \begin{pmatrix} x \\ x' \end{pmatrix} \cdot \begin{pmatrix} 5 & -7 \\ -7 & 10 \end{pmatrix} \begin{pmatrix} x \\ x' \end{pmatrix}$, then $B((x, x'), (y, y')) = \begin{pmatrix} x \\ x' \end{pmatrix} \cdot \begin{pmatrix} 5 & -7 \\ -7 & 10 \end{pmatrix} \begin{pmatrix} y \\ y' \end{pmatrix} = 5xy - 7xy' - 7x'y + 10x'y'$.

The polynomials $Q(\mathbf{x})$ and $B(\mathbf{x}, \mathbf{y})$ determine each other: $Q(\mathbf{x}) = B(\mathbf{x}, \mathbf{x})$ and

$$\begin{aligned} Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}) &= (\mathbf{x} + \mathbf{y}) \cdot A(\mathbf{x} + \mathbf{y}) - \mathbf{x} \cdot A\mathbf{x} - \mathbf{y} \cdot A\mathbf{y} \\ &= \mathbf{x} \cdot A\mathbf{y} + \mathbf{y} \cdot A\mathbf{x} \\ &= B(\mathbf{x}, \mathbf{y}) + B(\mathbf{y}, \mathbf{x}) \\ &= 2B(\mathbf{x}, \mathbf{y}), \end{aligned}$$

²The term “form” in quadratic form and bilinear form means a homogeneous function. Compare with differential forms.

so $B(\mathbf{x}, \mathbf{y}) = \frac{1}{2}(Q(\mathbf{x} + \mathbf{y}) - Q(\mathbf{x}) - Q(\mathbf{y}))$. This is a higher-dimensional generalization of the identity $xy = \frac{1}{2}((x + y)^2 - x^2 - y^2)$.

Exercises.

1. Show $M\mathbf{x} \cdot \mathbf{y} = \mathbf{x} \cdot M^\top \mathbf{y}$ for all $k \times k$ matrices M and vectors $\mathbf{x}$ and $\mathbf{y}$ with k components.
2. Check $ax^2 + bxy + cy^2 = \begin{pmatrix} x \\ y \end{pmatrix} \cdot \begin{pmatrix} a & b \\ b/2 & c \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$. Comparing that with Example 1.3, the matrices $\begin{pmatrix} a & b/2 \\ b/2 & c \end{pmatrix}$ and $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$ are not equal when $b \neq 0$. Why doesn't this contradict the uniqueness of A in Theorem 1.1?
3. Show Theorem 1.1 has a counterexample over every field K characteristic 2: there is *no* 2×2 symmetric matrix A with entries in K such that $x^2 + xy + y^2 = \mathbf{x} \cdot A\mathbf{x}$.

2. DEFINITENESS

The sum of squares quadratic form $x_1^2 + x_2^2 + \cdots + x_k^2$ only has positive values on $\mathbf{R}^k$ except at $\mathbf{0} = (0, \dots, 0)$, where it is 0. Here is a generalization of this example.

Definition 2.1. A k -dimensional quadratic form $Q(\mathbf{x})$ over $\mathbf{R}$ is called *positive-definite* if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{R}^k$ and $Q(\mathbf{x}) = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Example 2.2. Set $Q(x, y) = x^2 - xy + y^2$. By completing the square,

$$x^2 - xy + y^2 = \left(x - \frac{y}{2}\right)^2 + 3\frac{y^2}{4} = x'^2 + 3y'^2$$

where $x' = x - y/2$ and $y' = y/2$. Since $x'^2 + 3y'^2 \geq 0$ for all x' and y' and $x'^2 + 3y'^2 = 0$ only when $x' = 0$ and $y' = 0$, which is equivalent to $x = 0$ and $y = 0$,

Example 2.3. Let $\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \cdots + x_k^2} = \sqrt{\mathbf{x} \cdot \mathbf{x}}$ be the Euclidean norm on $\mathbf{R}^k$. For an invertible real $k \times k$ matrix M , set $Q(\mathbf{x}) = \|M\mathbf{x}\|^2 = M\mathbf{x} \cdot M\mathbf{x}$. The vector $M\mathbf{x}$ has components that are linear combinations of the components of $\mathbf{x}$, so $M\mathbf{x} \cdot M\mathbf{x}$ is a homogeneous degree 2 polynomial in the components of $\mathbf{x}$: it is a quadratic form. From its definition, $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, and if $Q(\mathbf{x}) = 0$ then $\|M\mathbf{x}\| = 0$, so $M\mathbf{x} = \mathbf{0}$ and thus $\mathbf{x} = \mathbf{0}$ since M is invertible. This shows $Q(\mathbf{x})$ is positive-definite.

The next theorem says that the previous example of a construction of positive-definite quadratic forms explains *all* positive-definite quadratic forms. The key tool in the proof is the spectral theorem for real symmetric matrices.

Theorem 2.4. Let $Q(\mathbf{x})$ be a k -dimensional quadratic form over $\mathbf{R}$. If Q is positive-definite then there is an $k \times k$ invertible real symmetric matrix M such that $Q(\mathbf{x}) = \|M\mathbf{x}\|^2 = M\mathbf{x} \cdot M\mathbf{x}$ for all $\mathbf{x} \in \mathbf{R}^k$.

Proof. Since $Q(\mathbf{x})$ is a quadratic form over $\mathbf{R}$, by Theorem 1.1 we can write $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ for a symmetric $k \times k$ real matrix A , and A is invertible: if A were *not* invertible there would be a nonzero $\mathbf{x} \in \mathbf{R}^k$ such that $A\mathbf{x} = \mathbf{0}$. Then $Q(\mathbf{x}) = \mathbf{x} \cdot \mathbf{0} = 0$, which by positive-definiteness implies $\mathbf{x} = \mathbf{0}$: contradiction.

The spectral theorem for real symmetric matrices says that every such matrix has a basis of orthogonal eigenvectors. Thus A has a basis $\mathbf{v}_1, \dots, \mathbf{v}_k$ where $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ and $i \neq j$ then $\mathbf{v}_i \perp \mathbf{v}_j$, so $\mathbf{v}_i \cdot \mathbf{v}_j = 0$. We can scale the $\mathbf{v}_i$'s so they all have length 1. Then $\mathbf{v}_1, \dots, \mathbf{v}_k$ is

an orthonormal basis of $\mathbf{R}^k$, not just an orthogonal basis: they are mutually perpendicular with length 1, not just mutually perpendicular.

From the formula for $Q(\mathbf{x})$ as the dot product $\mathbf{x} \cdot A\mathbf{x}$, $Q(\mathbf{v}_i) = \mathbf{v}_i \cdot A\mathbf{v}_i = \lambda_i(\mathbf{v}_i \cdot \mathbf{v}_i)$ for all i . The dot product $\mathbf{v}_i \cdot \mathbf{v}_i$ is $\|\mathbf{v}_i\|^2 = 1$ and $Q(\mathbf{v}_i) > 0$ since Q is positive-definite. Then $\lambda_i = Q(\mathbf{v}_i)/(\mathbf{v}_i \cdot \mathbf{v}_i) = Q(\mathbf{v}_i)$, so all the eigenvalues of A are positive when $\mathbf{x} \cdot A\mathbf{x}$ is positive-definite.

Let C be the change of basis matrix from the standard basis $\mathbf{e}_1, \dots, \mathbf{e}_k$ of $\mathbf{R}^k$ to the eigenbasis $\mathbf{v}_1, \dots, \mathbf{v}_k$ of A : $C\mathbf{e}_i = \mathbf{v}_i$ for all i , so the columns of C are $\mathbf{v}_1, \dots, \mathbf{v}_k$. Since the columns of C are an orthonormal basis of $\mathbf{R}^k$, we have dot product formulas $\mathbf{v}_i \cdot \mathbf{v}_j = 1$ if $i = j$ and 0 if $i \neq j$. That implies $C^\top C = I_k$, so $C^{-1} = C^\top$.

Let $D = \text{diag}(\lambda_1, \dots, \lambda_k)$ be the diagonal matrix whose i th diagonal entry is the eigenvalue of A for $\mathbf{v}_i$: $D\mathbf{e}_i = \lambda_i\mathbf{e}_i$. Then $AC = CD$ since the two sides here are linear on $\mathbf{R}^k$ and agree on the basis $\mathbf{e}_1, \dots, \mathbf{e}_k$: the common value of AC and CD at each $\mathbf{e}_i$ is $\lambda_i\mathbf{v}_i$ (check!). From $AC = CD$, we have $A = CDC^{-1}$. The eigenvalues of A are all positive, so write $\lambda_i = s_i^2$ where $s_i > 0$. Then $D = S^2$ where $S = \text{diag}(s_1, \dots, s_k)$, so $A = CDC^{-1} = CS^2C^{-1} = (CSC^{-1})^2 = (CSC^\top)^2$ and $CSC^\top$ is symmetric: $(CSC^\top)^\top = CS^\top C^\top = CSC^\top$ since S , being diagonal, is its own transpose.

Set $M = CSC^{-1} = CSC^\top$, so $M = M^\top$ and $A = M^2$, so A being invertible implies M is invertible. This implies

$$Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x} = \mathbf{x} \cdot M^2\mathbf{x} = M^\top \mathbf{x} \cdot M\mathbf{x} = M\mathbf{x} \cdot M\mathbf{x} = \|M\mathbf{x}\|^2,$$

which is what we wanted to show. $\square$

Corollary 2.5. *When $Q(\mathbf{x})$ is a k -dimensional positive-definite quadratic form over $\mathbf{R}$, its level sets $\{\mathbf{x} : Q(\mathbf{x}) = c\}$ for each $c \geq 0$ are bounded. In particular, the equation $Q(\mathbf{x}) = c$ has finitely many solutions in $\mathbf{Z}^k$.*

Proof. By Theorem 2.4, there is a $k \times k$ invertible real (symmetric) matrix M such that $Q(\mathbf{x}) = \|M\mathbf{x}\|^2$ for all $\mathbf{x}$ in $\mathbf{R}^k$. Then $Q(\mathbf{x}) = c$ is the same as $\|M\mathbf{x}\| = \sqrt{c}$. The set $S := \{\mathbf{v} : \|\mathbf{v}\| = \sqrt{c}\}$ is a sphere in $\mathbf{R}^k$ of radius $\sqrt{c}$ and

$$Q(\mathbf{x}) = c \iff \mathbf{x} \in M^{-1}(S_{\sqrt{c}}).$$

The sphere $S_{\sqrt{c}}$ is compact and since $M^{-1} : \mathbf{R}^k \rightarrow \mathbf{R}^k$ is linear it is continuous (Exercise 2.3), so the image $M^{-1}(S_{\sqrt{c}})$ in $\mathbf{R}^k$ is compact. Thus this image is bounded. $\square$

A quadratic form over $\mathbf{R}$ that is not positive definite (and not negative definite, which is the same as positive definite except positivity of values away from the origin are replaced by negativity) can have unbounded level sets, so these may contain infinitely many integral points. Consider $Q(x, y) = x^2 - 2y^2$ and the 1-level set $\{(x, y) \in \mathbf{R}^2 : x^2 - 2y^2 = 1\}$, which has infinitely many integral points since Pell's equation has infinitely many integral solutions.

Definition 2.6. For a k -dimensional integral positive-definite quadratic form Q , its *representation numbers* are $r_Q(n) := |\{\mathbf{x} \in \mathbf{Z}^k : Q(\mathbf{x}) = n\}|$ for integers $n \geq 0$.

When $Q(x_1, \dots, x_k) = x_1^2 + x_2^2 + \dots + x_k^2$, the numbers $r_Q(n)$ are written as $r_k(n)$: this is the number of k -tuples of integers whose squares add up to n .

The table below lists the initial values of $r_Q(n)$ for 6 positive-definite quadratic forms.³

³In the rows for $Q(x, y) = x^2 + 3y^2$ and $Q(x, y) = x^2 - xy + y^2$ we see the entries are both 0 and both nonzero for the same n . This is a general pattern for these two quadratic forms.

n	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$r_{x^2+y^2}(n)$	1	4	4	0	4	8	0	0	4	4	8	0	0	8	0	0
$r_{x^2+3y^2}(n)$	1	2	0	2	6	0	0	4	0	2	0	0	6	4	0	0
$r_{x^2-xy+y^2}(n)$	1	6	0	6	6	0	0	12	0	6	0	0	6	12	0	0
$r_{x^2+5y^2}(n)$	1	2	0	0	2	2	4	0	0	6	0	0	0	0	4	0
$r_{2x^2-2xy+3y^2}(n)$	1	0	2	4	0	0	0	4	2	0	2	0	4	0	0	4
$r_{x^2+y^2+z^2}(n)$	1	6	12	8	6	24	24	0	12	30	24	24	8	24	48	0

An important way to study the representation numbers $r_Q(n)$ as n varies is through analytic properties of a suitable generating function for this sequence. Here are two such generating functions.

Zeta-function of Q : for complex s , set

$$\zeta_Q(s) = \sum_{\mathbf{x} \in \mathbf{Z}^k, \mathbf{x} \neq \mathbf{0}} \frac{1}{Q(\mathbf{x})^s} = \sum_{n \geq 1} \frac{r_Q(n)}{n^s},$$

where we get the second series from the first by collecting together all terms with the same denominator n : each $1/n^s$ appears in the series as often as we can solve $Q(\mathbf{x}) = n$ for $\mathbf{x}$ in $\mathbf{Z}^k$, and that is $r_Q(n)$ times. The series $\zeta_Q(s)$ has constant term $r_Q(1)$: the number $r_Q(0)$ doesn't appear. It turns out that $\zeta_Q(s)$ is absolutely convergent on the right half-plane $\{s : \operatorname{Re}(s) > k/2\}$, where it is analytic.

Example 2.7. Taking $k = 2$, let $Q(x, y) = x^2 + 5y^2$, so $\zeta_Q(s) = 2 + 2/4^s + 2/5^s + 4/6^s + 6/9^s + 4/14^s + \dots$ for $\operatorname{Re}(s) > 2/2 = 1$. If we scale $\zeta_Q(s)$ so its constant term is 1, we get $(1/2)\zeta_Q(s) = 1 + 1/4^s + 1/5^s + 2/6^s + 3/9^s + 2/14^s + \dots$.

Example 2.8. Taking $k = 1$, let $Q(x) = x^2$, so $r_Q(n)$ is 2 if $n > 0$ is a perfect square and 0 otherwise. Thus $\zeta_Q(s) = \sum_{m \geq 1} 2/(m^2)^s = \sum_{m \geq 1} 2/m^{2s} = 2\zeta(2s)$ for $\operatorname{Re}(s) > 1/2$.

We'll see later that each $\zeta_Q(s)$ has a meromorphic continuation to $\mathbf{C}$ and a functional equation resembling that of the Riemann zeta-function. But unlike the Riemann zeta-function, $\zeta_Q(s)$ usually does *not* have an Euler product (after we scale it to make the constant term 1) or satisfy a Riemann hypothesis.

Theta-function of Q : for τ in the upper half-plane $\mathfrak{h} = \{z = x + iy : x, y \in \mathbf{R}, y > 0\}$, set

$$\theta_Q(\tau) = \sum_{\mathbf{x} \in \mathbf{Z}^k} e^{\pi i Q(\mathbf{x})\tau} = \sum_{n \geq 0} r_Q(n) e^{\pi i n \tau},$$

where we get the second series from the first by collecting together all terms with the same exponential term: each $e^{\pi i n \tau}$ for $n \geq 0$ occurs in $\theta_Q(\tau)$ as often as we can solve $Q(\mathbf{x}) = n$ for $\mathbf{x}$ in $\mathbf{Z}^k$.

Writing $\tau = x + iy$ with $y > 0$, $|e^{\pi i n \tau}| = |e^{\pi i n x} e^{-\pi n y}| = e^{-\pi n y}$ gets small when n gets big. The series $\theta_Q(\tau)$ is convergent on all of $\mathfrak{h}$, where it is analytic. It turns out to be a modular form for a suitable subgroup of $\operatorname{SL}_2(\mathbf{Z})$, but usually is not a modular form for all of $\operatorname{SL}_2(\mathbf{Z})$.

Example 2.9. When $Q(x, y) = x^2 + 5y^2$,

$$\theta_Q(s) = 1 + 2e^{\pi i \tau} + 2e^{4\pi i \tau} + 2e^{5\pi i \tau} + 4e^{6\pi i \tau} + 6e^{9\pi i \tau} + 4e^{\pi i 14\tau} + \dots$$

Both $\zeta_Q(s)$ and $\theta_Q(\tau)$ depend on Q just through the numbers $r_Q(n)$ as n varies. When we make an invertible *integral* linear change of variables⁴ (like $(x, y) \mapsto (x + y, y)$, with inverse

⁴A linear change of variables is described by a matrix, and saying the change of variables is invertible over $\mathbf{Z}$ means the matrix is integral with determinant ± 1 .

$(x, y) \mapsto (x - y, y)$, but not $(x, y) \mapsto (x + y, x - y)$ with inverse $(x, y) \mapsto ((x + y)/2, (x - y)/2)$, where the inverse map isn't defined over $\mathbf{Z}$), a quadratic form Q turns into another quadratic form Q' with the same representation numbers, so $\zeta_Q = \zeta_{Q'}$ and $\theta_Q = \theta_{Q'}$. For example, $(x, y) \mapsto (x + y, y)$ turns $Q(x, y) = x^2 + y^2$ into $Q'(x, y) = (x + y)^2 + y^2 = x^2 + 2xy + 2y^2$, so $r_Q(n) = r_{Q'}(n)$ for all integers $n \geq 0$. Is a converse true? That is, if Q and Q' are two positive-definite integral quadratic forms of the same dimension and $r_Q(n) = r_{Q'}(n)$ for all $n \geq 0$, are Q and Q' linked by an invertible integral linear change of variables? The answer in general is *no*: in 1964, Milnor gave the first counterexamples in dimension 16. In 1992, Conway and Sloane found a counterexample in dimension 4 for the quadratic forms

$$Q(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 - xy - 4xz - 4xt - 4yt + zt$$

and

$$Q'(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 + 2xy + 2xz + 5xt - 3yz + 2yt.$$

Here $r_Q(n) = r_{Q'}(n)$ for all $n \geq 0$, but we can't turn Q into Q' or *vice versa* by an invertible integral linear change of variables.

Exercises.

- Let $f(x_1, \dots, x_k) \in \mathbf{R}[x_1, \dots, x_k]$ be homogeneous in k variables with *even* degree. Show that if $f(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{Z}^k$ then $f(\mathbf{x}) \geq 0$ for all $\mathbf{x} \in \mathbf{R}^k$; the converse is obvious. In particular, a k -dimensional quadratic form over $\mathbf{Z}$ that is nonnegative on $\mathbf{Z}^k$ is nonnegative on $\mathbf{R}^k$. (Homogeneity of f is important here: $6x^2 - 6x + 1$ is not homogeneous and is positive on all of $\mathbf{Z}$ but it is not positive on all of $\mathbf{R}$.)
- Show a binary quadratic form $ax^2 + bxy + cy^2$ is positive-definite if and only if $a > 0$ and $b^2 - 4ac < 0$, or equivalently $c > 0$ and $b^2 - 4ac < 0$. (Hint: complete the square.)
- When $L : \mathbf{R}^k \rightarrow \mathbf{R}^k$ is a linear transformation, show L is continuous. (Hint: look at the formulas for the component functions of L .) Your argument should be extendable to the case of linear transformations $\mathbf{R}^k \rightarrow \mathbf{R}^\ell$ where k and ℓ might be different: such functions are continuous. Can you show linear transformations between Euclidean spaces are uniformly continuous?
- Find an invertible *integral* linear change of variables that converts $5x^2 - 14xy + 10y^2$ into $x^2 + y^2$.
- (Positive-definiteness and eigenvalues)
 - Let $Q(\mathbf{x}) = \mathbf{x} \cdot A\mathbf{x}$ be a quadratic form over $\mathbf{R}$, where A is a $k \times k$ real symmetric matrix. When A has eigenvalues $\lambda_1, \dots, \lambda_k$, which are necessarily all real, use the spectral theorem to show there is a change of basis of $\mathbf{R}^k$ that makes Q take on the form $\sum_{i=1}^k \lambda_i x_i^2$. In particular, Q is positive-definite if and only if all the eigenvalues of A are positive. (Note: for $k > 1$, Q can have many diagonal forms $\sum_{i=1}^k c_i x_i^2$ without the c_i 's being eigenvalues of A . That there is a diagonal form for Q with coefficients that are eigenvalues of A is merely a property of those eigenvalues and not a characterization of them.)
 - Use Descartes' rule of signs to show all the eigenvalues of A are positive if and only if the coefficients of the characteristic polynomial of A have alternating signs. This allows you to check a quadratic form over $\mathbf{R}$ is positive-definite without having to calculate its eigenvalues or put Q into a diagonal form.

c) Check the 4-dimensional quadratic form

$$Q(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 - xy - 4xz - 4xt - 4yt + zt$$

is $\mathbf{x} \cdot A\mathbf{x}$ where

$$A = \begin{pmatrix} 2 & -1/2 & -2 & -2 \\ -1/2 & 4 & 0 & -2 \\ -2 & 0 & 6 & 1/2 \\ -2 & -2 & 0 & 8 \end{pmatrix}$$

and calculate the characteristic polynomial of A to verify that Q is positive-definite using part (b).

d) Do the same as part (c) for the quadratic form

$$Q'(x, y, z, t) = 2x^2 + 4y^2 + 6z^2 + 8t^2 + 2xy + 2xz + 5xt - 3yz + 2yt.$$

As a first step, work out the 4×4 symmetric matrix A' such that $Q'(\mathbf{x}) = \mathbf{x} \cdot A'\mathbf{x}$.

3. TEMPLATE FOR ANALYTIC CONTINUATION: THE RIEMANN ZETA-FUNCTION

For a k -dimensional positive-definite integral quadratic form $Q(\mathbf{x})$, showing that the series defining $\zeta_Q(s)$ converges absolutely on a right half-plane and has a meromorphic continuation to $\mathbf{C}$ that satisfies a functional equation will be based on one of the methods used to prove similar results for the Riemann zeta-function $\zeta(s) = \sum_{n \geq 1} 1/n^s$, so in this section we will review the story for $\zeta(s)$.

Using the integral test, $\sum_{n \geq 1} 1/n^s$ converges absolutely on the right half-plane $\operatorname{Re}(s) > 1$ and it converges uniformly on compact subsets of that half-plane. Uniform convergence on compact subsets of $\operatorname{Re}(s) > 1$ implies $\zeta(s)$ is analytic on $\operatorname{Re}(s) > 1$. To extend $\zeta(s)$ to all of $\mathbf{C}$, we will use the Gamma-function $\Gamma(s)$. This is defined on the right half-plane $\operatorname{Re}(s) > 0$ by an integral depending on s :

$$\Gamma(s) = \int_0^\infty e^{-x} x^s \frac{dx}{x}.$$

For example, $\Gamma(1) = \int_0^\infty e^{-x} dx = 1$. It can be shown by a suitable theorem about differentiating under the integral sign that $\Gamma(s)$ is analytic on $\operatorname{Re}(s) > 0$.

Using integration by parts, we get

$$\Gamma(s+1) = s\Gamma(s),$$

so $\Gamma(s) = \Gamma(s+1)/s$. The right side, $\Gamma(s+1)/s$, makes sense for $\operatorname{Re}(s) > -1$ except for a simple pole at $s = 0$. Thus the equation $\Gamma(s) = \Gamma(s+1)/s$ extends $\Gamma(s)$ from $\operatorname{Re}(s) > 0$ to $\operatorname{Re}(s) > -1$, where it is analytic except for the simple pole at $s = 0$. Then for $\operatorname{Re}(s) > -1$,

$$\Gamma(s) = \frac{\Gamma(s+1)}{s} = \frac{\Gamma(s+2)/(s+1)}{s} = \frac{\Gamma(s+2)}{s(s+1)}$$

and the last ratio lets us extend $\Gamma(s)$ analytically to $\operatorname{Re}(s) > -2$ except for simple poles at $s = 0$ and $s = 1$. Continuing in this way, $\Gamma(s)$ extends analytically to $\mathbf{C}$ except for simple poles at $s = 0$ and all negative integers. (See Exercise 3.3 for the residues of $\Gamma(s)$ at its poles.)

Returning to $\zeta(s)$, for $\operatorname{Re}(s) > 1$ define the *completed zeta-function* to be

$$Z(s) = \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s).$$

We will show $Z(s)$ has an analytic continuation to $\mathbf{C}$ except for simple poles at $s = 0$ and 1 with respective residues -1 and 1 .

To begin, for $\operatorname{Re}(s) > 0$

$$(3.1) \quad \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \frac{1}{n^s} = \int_0^\infty e^{-x} \left(\frac{x}{\pi n^2}\right)^{s/2} \frac{dx}{x} = \int_0^\infty e^{-\pi n^2 y} y^{s/2} \frac{dy}{y},$$

where we made the multiplicative change of variables $y = x/(\pi n^2)$, so $dy/y = dx/x$.

For $\operatorname{Re}(s) > 1$, the sum of (3.1) over all positive integers n converges to $\pi^{-s/2} \Gamma(s/2) \zeta(s)$, leading to

$$(3.2) \quad Z(s) := \pi^{-s/2} \Gamma\left(\frac{s}{2}\right) \zeta(s) = \sum_{n \geq 1} \int_0^\infty e^{-\pi n^2 y} y^{s/2} \frac{dy}{y} = \int_0^\infty \omega(y) y^{s/2} \frac{dy}{y},$$

where

$$\omega(y) := \sum_{n \geq 1} e^{-\pi n^2 y}$$

and the interchange of sum and integral can be justified by the Dominated Convergence Theorem.

The main subtle aspect of the integral representation of $Z(s)$ at the end of (3.2) is that the integral has lower bound $x = 0$. Indeed, for $c > 0$,

$$\int_c^\infty \omega(y) y^s dy = \int_c^\infty \left(\sum_{n \geq 1} e^{-\pi n^2 y} \right) y^s dy$$

converges absolutely for all s in $\mathbf{C}$ and is entire. That is false for $c = 0$. We will make progress in understanding $Z(s)$ by replacing $\int_0^\infty$ at the end of (3.2) with $\int_1^\infty$.

For $y > 0$, set

$$(3.3) \quad \Theta(y) := \sum_{n \in \mathbf{Z}} e^{-\pi n^2 y} = 1 + 2 \sum_{n \geq 1} e^{-\pi n^2 y} = 1 + 2\omega(y).$$

Theorem 3.1. *For positive y , $\Theta(1/y) = \sqrt{y} \Theta(y)$. Equivalently,*

$$\sum_{n \in \mathbf{Z}} e^{-\pi n^2 / y} = \sqrt{y} \sum_{n \in \mathbf{Z}} e^{-\pi n^2 y}.$$

From the viewpoint of numerical analysis, Theorem 3.1 is useful for computing $\Theta(y)$ when y is very small, since in that case the series for $\Theta(y)$ converges slowly while the series for $\Theta(1/y)$ converges rapidly since $1/y$ is large. The table below illustrates Theorem 3.1.

y	2	3	4	5
$\Theta(y)$	1.003734...	1.000161...	1.000006...	1.000000...
$\Theta(1/y)$	1.419495...	1.732330...	2.000013...	2.236068...

The proof of Theorem 3.1 will be postponed to the end of this section in order to see first how theorem helps us.

Returning to $Z(s)$, break up the integral over $(0, \infty)$ in (3.2) at $y = 1$, since this is the symmetry point for $y \leftrightarrow 1/y$. Replacing y with $1/y$ converts $(0, 1]$ to $[1, \infty)$ and converts dy/y to $-dy/y$, so

$$(3.4) \quad Z(s) = \int_1^\infty \omega\left(\frac{1}{y}\right) y^{-s/2} \frac{dy}{y} + \int_1^\infty \omega(y) y^{s/2} \frac{dy}{y}.$$

For $y \geq 1$, $\omega(y) = e^{-\pi y} + e^{-4\pi y} + \dots$ has exponential decay as $y \rightarrow \infty$, so the second integral in (3.4) converges for all s in $\mathbf{C}$. By Theorem 3.1, the first integral in (3.4) is

$$\begin{aligned} \int_1^\infty \frac{1}{2}(\sqrt{y}(1 + 2\omega(y)) - 1)y^{-s/2} \frac{dy}{y} &= \int_1^\infty \omega(y)y^{(1-s)/2} \frac{dy}{y} + \\ &\quad \frac{1}{2} \int_1^\infty (\sqrt{y} - 1)y^{-s/2} \frac{dy}{y} \\ &= \int_1^\infty \omega(y)y^{(1-s)/2} \frac{dy}{y} + \frac{1}{s-1} - \frac{1}{s} \end{aligned}$$

since $\int_1^\infty dy/y^z = 1/(z-1)$ if $\operatorname{Re}(z) > 1$ (Exercise 3.4(a)). Comparing this integral to the second one in (3.4), $1-s$ appears in place of s , so the integral with $1-s$ converges for all $s \in \mathbf{C}$.

We have obtained an analytic formula for $Z(s)$ that is valid for all complex s :

$$\begin{aligned} Z(s) &= \int_1^\infty \omega(y)(y^{s/2} + y^{(1-s)/2}) \frac{dy}{y} - \frac{1}{s} - \frac{1}{1-s} \\ (3.5) \quad &= \int_1^\infty \frac{1}{2}(\Theta(y) - 1)(y^{s/2} + y^{(1-s)/2}) \frac{dy}{y} - \frac{1}{s} - \frac{1}{1-s}. \end{aligned}$$

This formula is unchanged if s is replaced with $1-s$, which proves $Z(s) = Z(1-s)$. By (3.5) we get an analytic continuation of $Z(s)$ to $\mathbf{C}$ except for simple poles at $s = 0$ and $s = 1$ with respective residues -1 and 1 (The polar term at $s = 1$ is $-1/(1-s) = 1/(s-1)$.)

Let's summarize what we have found.

Theorem 3.2. *The function $Z(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$ extends from $\operatorname{Re}(s) > 1$ to a meromorphic function on $\mathbf{C}$ with simple poles at 0 and 1, having respective residues -1 and 1. It satisfies the functional equation $Z(1-s) = Z(s)$.*

Let's convert this into a result about $\zeta(s)$.

Corollary 3.3. *The Riemann zeta-function has an analytic continuation to $\mathbf{C}$ except for a simple pole at $s = 1$ with residue 1.*

Proof. For each $s \in \mathbf{C}$, set $\zeta(s) = \pi^{s/2}Z(s)/\Gamma(s/2)$. This comes from the definition of $Z(s)$ in terms of $\zeta(s)$ on $\operatorname{Re}(s) > 1$, so it provides a meromorphic continuation of $\zeta(s)$ to $\mathbf{C}$. The function $Z(s)$ is analytic on $\mathbf{C} - \{0, 1\}$ and it can be shown that $1/\Gamma(s)$ is an entire function, so $1/\Gamma(s/2)$ is an entire function as well. Thus the only possible poles of $\zeta(s)$ could be at $s = 0$ or 1. At $s = 1$, we saw $Z(s)$ has a simple pole with residue 1, so $\zeta(s)$ at $s = 1$ has a simple pole with residue $\pi^{1/2}/\Gamma(1/2)$. It can be shown that $\Gamma(1/2) = \sqrt{\pi}$, so $\zeta(s)$ at $s = 1$ has residue 1. At $s = 0$, the simple poles of $Z(s)$ and $\Gamma(s/2)$ cancel, making $\zeta(s)$ analytic there with $\zeta(0) \neq 0$. (In fact, $\zeta(0) = -1/2$.) $\square$

Corollary 3.4 (Asymmetric Functional Equation). *For every s ,*

$$\zeta(1-s) = 2(2\pi)^{-s}\Gamma(s) \cos\left(\frac{\pi s}{2}\right) \zeta(s).$$

Proof. From $Z(1-s) = Z(s)$,

$$\zeta(1-s) = \frac{\pi^{-s/2}\Gamma(s/2)}{\pi^{-(1-s)/2}\Gamma((1-s)/2)} \zeta(s) = \pi^{1/2-s} \frac{\Gamma(s/2)}{\Gamma((1-s)/2)} \zeta(s).$$

The Gamma function satisfies two important identities that are called the reflection and duplication formulas:

$$(3.6) \quad \Gamma(s)\Gamma(1-s) = \frac{\pi}{\sin(\pi s)}, \quad \Gamma\left(\frac{s}{2}\right)\Gamma\left(\frac{s+1}{2}\right) = \frac{\sqrt{\pi}}{2^{s-1}}\Gamma(s).$$

Using these, check as an exercise that

$$(3.7) \quad \frac{\Gamma(s/2)}{\Gamma((1-s)/2)} = \frac{2^{1-s}}{\sqrt{\pi}} \cos\left(\frac{\pi}{2}s\right)\Gamma(s).$$

Thus

$$\zeta(1-s) = \pi^{1/2-s} \frac{2^{1-s}}{\sqrt{\pi}} \cos\left(\frac{\pi}{2}s\right)\Gamma(s)\zeta(s) = 2(2\pi)^{-s}\Gamma(s) \cos\left(\frac{\pi}{2}s\right)\zeta(s). \quad \square$$

As the last item in this section, we prove the identity

$$(3.8) \quad \Theta\left(\frac{1}{y}\right) = \sqrt{y}\Theta(y),$$

where $y > 0$ and $\Theta(y) = \sum_{n \in \mathbf{Z}} e^{-\pi n^2 y} = 1 + 2\omega(y)$. The tools that lie behind (3.8) are the Fourier transform and the Poisson summation formula.

Definition 3.5. The *Fourier transform* of a continuous function $f: \mathbf{R} \rightarrow \mathbf{C}$ such that $\int_{\mathbf{R}} |f(x)| dx < \infty$ is the function $\hat{f}: \mathbf{R} \rightarrow \mathbf{C}$ that is defined by

$$\hat{f}(y) := \int_{-\infty}^{\infty} f(x) e^{-2\pi i x y} dx.$$

Example 3.6. If $f(x) = e^{-|x|}$, then $\hat{f}(y) = 2/(1 + 4\pi^2 y^2)$. See Exercise 3.3.

Theorem 3.7. The function $f(x) = e^{-\pi x^2}$ equals its own Fourier transform: $\hat{f}(y) = e^{-\pi y^2}$.

Proof. Rewrite $\hat{f}(y) = e^{-\pi y^2}$ as $1 = e^{\pi y^2} \hat{f}(y)$, which is the same as

$$(3.9) \quad 1 = \int_{-\infty}^{\infty} e^{-\pi(x+iy)^2} dx = \int_{\{x+iy: x \in \mathbf{R}\}} e^{-\pi w^2} dw.$$

This says the integral over $\mathbf{R}$ should be independent of y , and we'll prove that.

That (3.9) holds at $y = 0$ says $\int_{\mathbf{R}} e^{-\pi x^2} dx = 1$. This is equivalent to $\frac{1}{\sqrt{2\pi}} \int_{\mathbf{R}} e^{-x^2/2} dx = 1$. See <https://kconrad.math.uconn.edu/blurbs/analysis/gaussianintegral.pdf> for multiple proofs.

To prove (3.9) for $y > 0$, take a rectangular contour with vertices at R , $R+iy$, $-R+iy$, and $-R$, as shown in Figure 1. The integral of $e^{-\pi w^2}$ around this contour (in either direction) is 0 by Cauchy's theorem.

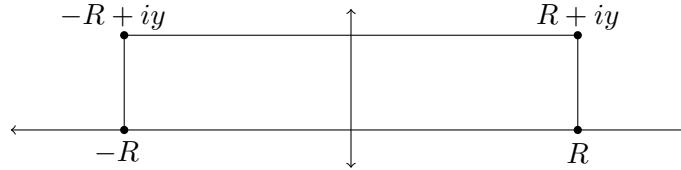


FIGURE 1. Contour for integral of $e^{-\pi w^2}$.

Check by estimates that the integrals on the vertical edges tend to 0 as $R \rightarrow \infty$ (Exercise 3.8(a)), so the integrals on the horizontal edges, if oriented *in the same direction*, agree in the limit as $R \rightarrow \infty$. Thus $\int_{-\infty}^{\infty} e^{-\pi(x+iy)^2} dx = \int_{-\infty}^{\infty} e^{-\pi x^2} dx$, and the second integral is 1, so the first is as well.

To prove $\int_{-\infty}^{\infty} e^{-\pi(x+iy)^2} dx = \int_{-\infty}^{\infty} e^{-\pi x^2} dx$ when $y < 0$, a similar argument could be made using a rectangular contour extending into the lower half-plane or you could take complex conjugates of both sides of the proved equation $\int_{-\infty}^{\infty} e^{-\pi(x+iy)^2} dx = \int_{-\infty}^{\infty} e^{-\pi x^2} dx$ for $y > 0$ to get the same result with $-y$ in place of y . $\square$

When $f = \hat{f}$ we say f is *self-dual*. The function $e^{-\pi x^2}$ is not the only self-dual function (up to scaling): see Exercise 3.8(b) for another one.

The *Poisson summation formula* on $\mathbf{R}$ says that for “nice” functions $h: \mathbf{R} \rightarrow \mathbf{C}$,

$$(3.10) \quad \sum_{n \in \mathbf{Z}} h(n) = \sum_{n \in \mathbf{Z}} \hat{h}(n).$$

An example of what “nice” can mean here is that h lies in the Schwartz space on $\mathbf{R}$, which is a space of rapidly decreasing functions $\mathbf{R} \rightarrow \mathbf{C}$ that is preserved by many linear operations in analysis, such as differentiation and the Fourier transform.

In (3.10), replace $h(x)$ by $g(x) := h(ax)$ for a nonzero constant a . The Fourier transform of g differs from that for h by a scaling factor:

$$(3.11) \quad g(x) = h(ax) \implies \hat{g}(y) = \int_{-\infty}^{\infty} h(ax) e^{-2\pi i xy} dx = \frac{1}{|a|} \hat{h}\left(\frac{y}{a}\right).$$

(To derive this, treat $a > 0$ and $a < 0$ separately.)

Poisson summation for $g(x) = h(ax)$ takes the form

$$(3.12) \quad \sum_{n \in \mathbf{Z}} h(an) = \frac{1}{|a|} \sum_{n \in \mathbf{Z}} \hat{h}\left(\frac{n}{a}\right).$$

Using $h(x) = e^{-\pi x^2}$ and $a = \sqrt{t}$ here, where $t > 0$, $\hat{h} = h$ and we get

$$\sum_{n \in \mathbf{Z}} e^{-\pi n^2 t} = \frac{1}{\sqrt{t}} \sum_{n \in \mathbf{Z}} e^{-\pi n^2 / t},$$

so $\Theta(t) = (1/\sqrt{t})\Theta(1/t)$, which is (3.8).

Exercises.

1. Show the series $\zeta(s) = \sum_{n \geq 1} 1/n^s$ converges uniformly on right half-planes $\operatorname{Re}(s) \geq 1 + \varepsilon$, where $\varepsilon > 0$. (Since every compact subset of $\operatorname{Re}(s) > 1$ is contained in some half-plane $\operatorname{Re}(s) \geq 1 + \varepsilon$, this shows $\zeta(s)$ converges uniformly on compact subsets of $\operatorname{Re}(s) > 1$.)
2. Show for $\operatorname{Re}(s) > 0$ that the integral $\Gamma(s) = \int_0^{\infty} e^{-x} x^s dx/x$ is absolutely convergent, meaning $\int_0^{\infty} e^{-x} x^a dx/x$ converges for all positive numbers a . The only subtlety is behavior of the integral near $x = 0$: pick $c > 0$ and consider separately absolute convergence of $\int_0^c e^{-x} x^a dx/x$ and $\int_c^{\infty} e^{-x} x^a dx/x$.
3. For an integer $n \geq 0$, show $\Gamma(s)$ has residue $(-1)^n/n!$.
4. (A Fourier transform calculation.)
 - a) For $z \in \mathbf{C}$, show $\int_1^{\infty} du/u^z = 1/(z-1)$ for $\operatorname{Re}(z) > 1$.
 - b) For $z \in \mathbf{C}$, show $\int_0^{\infty} e^{-xz} dx = 1/z$ for $\operatorname{Re}(z) > 0$.

- c) Use part (b) to show the function $f(x) = e^{-|x|}$ for $x \in \mathbf{R}$ has Fourier transform $\widehat{f}(y) = 2/(1 + 4\pi^2 y^2)$.
5. For $c > 0$, show the Fourier transform of e^{-cx^2} is $\sqrt{\pi/c} e^{-(\pi^2/c)y^2}$. (This says a highly peaked Gaussian has a Fourier transform that is spread out, and *vice versa*. One mathematical expression of the Heisenberg uncertainty relation from physics is that a function and its Fourier transform, other than 0, can't both be highly localized.)
6. When $g(x) = e^{2\pi i ax} f(x + b)$, show $\widehat{g}(y) = e^{2\pi i b(y-a)} \widehat{f}(y - a)$.
7. Use the duplication and reflection formulas for $\Gamma(s)$ in (3.6) to derive (3.7).
8. (Self-dual functions)
- a) In the proof of Theorem 3.7, check the integrals on the vertical edges in Figure 1 tend to 0 as $R \rightarrow \infty$.
 - b) Show the function $f(x) = 1/(e^x + e^{-x})$ is self-dual: it is equal to its own Fourier transform. (Up to a scaling factor of 2, $f(x)$ is the hyperbolic secant function.) Mimic the method used to show $e^{-\pi x^2}$ is self-dual.